

Yarncrawler: World-State Reconstruction as Constraint Closure over a Generative Field

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Abstract

This paper introduces Yarncrawler as a framework for reconstructing physical world-states from multi-modal sensor observations under continuous dynamical constraint. The central object is a consistency operator $\mathcal{C}$ acting on the quotient configuration space $\mathcal{X}/\sim$ of coupled Relativistic Scalar-Vector Plenum field triples $(\Phi, \mathbf{v}, S)$. Each sensor modality defines a projection operator $\Pi_i : \mathcal{X} \rightarrow \mathcal{Y}_i$; each observation y_i constrains the set of field configurations consistent with what the sensor reports. The consistency operator searches for the unique admissible configuration whose forward projections are compatible with all observations simultaneously.

The main result—the Yarncrawler Identifiability Theorem—establishes that $\mathcal{C}$ has a unique fixed point if and only if the induced map $\tilde{\Pi} : \mathcal{X}/\sim \rightarrow \prod_i \mathcal{Y}_i$ is injective on the feasible set $\mathcal{F} = \Omega_{\text{obs}} \cap \Gamma$. The Dynamical Disambiguation Theorem extends this to realistic sensing regimes where static observations underdetermine the world-state: trajectory fragments constrain the tangent bundle of the admissible manifold, removing degrees of freedom that no static projection can touch.

The paper further establishes that the variational consistency framework and the sheaf-theoretic repair framework—in which sensor modalities define local charts, observations define local sections, and closure corresponds to the existence of a unique global section—are formally equivalent. There exists a functor $\Psi : \mathbf{Proj}(\mathcal{X}) \rightarrow \mathbf{Sh}(\mathcal{X}/\sim)$ under which the consistency operator is the variational realization of the sheaf repair operator, projection degeneracy corresponds to non-trivial H^1 , and entropy collapse corresponds to cohomology trivialization. The RSVP entropy field S is identified as the local density of cohomological obstruction; its minimization under $\mathcal{C}$ is the physical expression of sheaf repair.

Deployment at planetary, regional, and site scales is unified by treating each scale as a different geometry of the feasible set under varying noise profiles. The framework is implemented as a sensor fusion architecture combining learned trajectory operators with physics-constrained forward models, applied

to coupled ecosystem and civilization dynamics calibrated from before-and-after observational pairs across modalities including infrared, LiDAR, RGB, audio, and photonic strip sensors.

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1 Introduction

1.1 The Problem

Given a family of projection operators $\{\Pi_i\}_{i \in \mathcal{I}}$ mapping an unknown world-state $[X] \in \mathcal{X}/\sim$ to observable quantities $y_i \in \mathcal{Y}_i$, and given a dynamical law governing how world-states evolve, the central question of this paper is:

Under what conditions does there exist a unique trajectory in the quotient configuration space $\mathcal{X}/\sim$ consistent with all projections simultaneously and admissible under the dynamics?

This is not a modeling question. It is an identifiability question. The paper does not propose a method for approximating world-states from sensor data—it establishes the conditions under which a unique world-state is *demand*ed by the data, and proves that when those conditions hold, a specific operator converges to it.

The setting is the following. The world-state is a coupled field triple $X = (\Phi, \mathbf{v}, S)$ —scalar density, vector flow, and entropy production—evolving over a compact spatial domain under the Relativistic Scalar-Vector Plenum (RSVP) field equations. The projections Π_i correspond to sensor modalities: infrared, LiDAR, RGB, audio, photonic strips. Each observation y_i constrains which field configurations are consistent with what the sensor reports. The consistency operator $\mathcal{C}$ searches for the configuration that minimizes total projection residual subject to physical admissibility. The dynamical admissibility set Γ restricts attention to configurations that lie on trajectories the world could have taken. The feasible set $\mathcal{F} = \Omega_{\text{obs}} \cap \Gamma$ is the intersection of all constraints. Closure—the condition that $|\mathcal{F}| = 1$ —is the condition that the world-state is uniquely determined.

1.2 The Central Result

The paper proves the following.

The problem of multi-modal consistency—finding a world-state whose forward projections are compatible with all available observations—is equivalent to the existence of a unique global section of a sheaf $\mathcal{S}_\Pi$ induced by the projection family over a cover of $\mathcal{X}/\sim$.

More precisely: the consistency operator $\mathcal{C}$ is the variational realization of the sheaf repair operator. Its fixed points are global sections. Its failure modes—projection degeneracy and regularizer flatness—are cohomological obstructions $H^1 \neq 0$ and degenerate H^0 . Adding sensors refines the cover and kills H^1 classes. Strengthening the dynamical constraint extends the cover into the temporal direction and kills

residual ambiguity that spatial sensing cannot reach. Closure is the condition $H^0 = 1$, $H^1 = 0$: the sheaf has exactly one global section.

This is the Yarncrawler Identifiability Theorem and its dynamical extension, proved in Sections 6 and 7 and their appendices.

1.3 The Duality

The framework has two faces, both present in this paper.

The *forward* direction takes a world-state as given and asks how a system maintains coherence under perturbation: how it detects tears in its semantic fabric, applies repair morphisms that trivialize Čech cocycles, exports entropy through stigmergic reinforcement, and sustains homeorhetic flows rather than static equilibria.

The *inverse* direction takes sensor observations as given and asks what world-state they structurally demand. Given what the sensors report, what field configuration is forced?

These are not complementary perspectives on the same phenomenon. They are dual realizations of the same mathematical object. Both directions study the sheaf $\mathcal{S}_\Pi$ over the quotient configuration space $\mathcal{X}/\sim$. The forward direction studies the internal dynamics of sections within $\mathcal{S}_\Pi$ —how they evolve, heal, and accumulate into global coherence. The inverse direction studies the conditions under which a unique global section exists and the operator that converges to it. The RSVP field triple $(\Phi, \mathbf{v}, S)$ parametrizes the stalks in both directions. The Yarncrawler framework is the unified theory of that sheaf and its dynamics.

The object common to both is neither the operator nor the sheaf taken separately. It is the equivalence between them—the functor $\Psi : \mathbf{Proj}(\mathcal{X}) \rightarrow \mathbf{Sh}(\mathcal{X}/\sim)$ proved in Section 2, which shows that a projection family and a sheaf over a configuration space are the same structure viewed from outside and inside respectively.

1.4 The Consequence

When closure is achieved—when $|\mathcal{F}| = 1$ —three things happen simultaneously that were previously three separate conditions.

The projection residuals collapse to within noise tolerance: $\|\Pi_i(X^*) - y_i\| \leq \epsilon_i$ for all i . The cohomology of the induced sheaf trivializes: $H^1(\mathcal{U}, \mathcal{S}_\Pi) = 0$ and $\dim H^0 = 1$. The RSVP entropy field S is minimized subject to admissibility: no further constraint can be satisfied that is not already satisfied.

These are not three criteria that must be checked independently. Under the Sheaf-Variational Equivalence of Section 2, they are one criterion expressed in three languages—variational, cohomological, and field-theoretic. The entropy of the RSVP

field at the fixed point is the local density of cohomological obstruction, and its minimization is the physical expression of sheaf repair.

The closed state $[X^*]$ is therefore not a best estimate. It is the unique trajectory consistent with all projections and admissible dynamics—the world-state that no alternative history can explain away.

1.5 Structure of the Paper

Section 2 establishes the Sheaf-Variational Equivalence. Sections 3 and 4 develop the prerequisite machinery. Section 5 defines the feasible set geometrically. Sections 6 and 7 contain the core theorems. Section 8 develops the inversion layer architecture. Section 9 defines closure operationally and proves equivalence of the three closure criteria. Section 10 unifies planetary, regional, and site-scale deployment. Section 11 positions the framework against prior work. Section 12 draws the epistemological consequence. Appendices A–E provide the functional-analytic foundations.

2 The Sheaf-Variational Equivalence

2.1 Two Descriptions of the Same Object

The Yarncrawler framework has two faces. The first describes a self-refactoring system that maintains coherence by repairing tears in its semantic fabric: local sections are parsed, inconsistencies are detected as Čech cocycles, and repair morphisms trivialize them until a global section exists. The second describes a consistency operator that searches for a field configuration whose forward projections are compatible with all available sensor observations simultaneously. These descriptions appear to operate at different levels of abstraction—one categorical and topological, the other variational and analytic. This section proves they are the same description.

The identification is not a metaphor. There is a precise functor between the category of sheaves of semantic sections over a cover and the category of projection families on the quotient configuration space $\mathcal{X}/\sim$. Under this functor, every concept in the sheaf-theoretic account has an exact counterpart in the variational account, and vice versa.

2.2 The Sheaf-Theoretic Account

Let $\mathcal{M}$ be a topological space equipped with an open cover $\mathcal{U} = \{U_i\}_{i \in \mathcal{I}}$. A presheaf $\mathcal{S}$ over $\mathcal{U}$ assigns to each U_i a category $\mathcal{S}(U_i)$ of local sections, and to each inclusion $U_j \subseteq U_i$ a restriction functor $\rho_{ij} : \mathcal{S}(U_i) \rightarrow \mathcal{S}(U_j)$.

Definition 2.1 (Gluing Condition). *A family of local sections $\{s_i \in \mathcal{S}(U_i)\}_{i \in \mathcal{I}}$ satisfies the gluing condition if for all $i, j \in \mathcal{I}$:*

$$\rho_{U_i, U_i \cap U_j}(s_i) = \rho_{U_j, U_i \cap U_j}(s_j) \quad \text{on } U_i \cap U_j.$$

A presheaf in which every compatible family of local sections admits a unique global section $s \in \mathcal{S}(\mathcal{M})$ with $s|_{U_i} = s_i$ is called a sheaf.

Definition 2.2 (Semantic Tear and Repair). *A semantic tear at overlap $U_i \cap U_j$ is a failure of the gluing condition. The discrepancy defines a Čech 1-cochain $c_{ij} = \rho_{ij}(s_j) - \rho_{ji}(s_i)$. A repair is a morphism in $\mathcal{S}(U_i \cap U_j)$ that trivializes c_{ij} . The space of unresolved tears is measured by $H^1(\mathcal{U}, \mathcal{S})$.*

Definition 2.3 (Sheaf Repair Operator). *The sheaf repair operator $\mathcal{R}_{\text{sheaf}}$ maps a family of inconsistent local sections to a minimally modified family satisfying the gluing condition, where minimality is measured by a seam cost functional $\mathcal{L}_{\text{seam}}$:*

$$\mathcal{R}_{\text{sheaf}}(\{s_i\}) = \arg \min_{\{s'_i\}} \sum_{i < j} \mathcal{L}_{\text{seam}}(s'_i|_{U_{ij}}, s'_j|_{U_{ij}}) + \sum_i \mathcal{L}_{\text{local}}(s'_i, s_i).$$

Global section existence corresponds to $\mathcal{L}_{\text{seam}} = 0$ at the optimum.

2.3 The Variational Account

The variational account centers on the consistency operator:

$$\mathcal{C}([X]) = \arg \min_{[X'] \in \mathcal{A}} \left[\sum_{i \in \mathcal{I}} \|\Pi_i(X') - y_i\|_{\mathcal{Y}_i}^2 + \mathcal{R}(X') \right]$$

where $\Pi_i : \mathcal{X} \rightarrow \mathcal{Y}_i$ are projection operators, y_i are observations, and $\mathcal{R}$ is the RSVP-consistent regularizer encoding admissibility. A fixed point $[X^*]$ satisfies $\Pi_i(X^*) \approx y_i$ for all i within tolerance ϵ_i .

2.4 The Dictionary

The following correspondence identifies every object in the sheaf-theoretic account with its exact counterpart in the variational account.

Sheaf-theoretic concept	Variational counterpart
Open set U_i in cover $\mathcal{U}$	Sensor modality i with space $\mathcal{Y}_i$
Local section $s_i \in \mathcal{S}(U_i)$	Observation $y_i \in \mathcal{Y}_i$
Restriction map ρ_{ij}	Projection operator Π_i
Gluing condition	Projection consistency $\Pi_i(X) = y_i$
Čech 1-cochain c_{ij}	Projection residual $\ \Pi_i(X) - y_i\ $
Coboundary (trivialized cocycle)	Residual driven to zero by $\mathcal{C}$
Global section $s \in \mathcal{S}(\mathcal{M})$	Fixed point $[X^*] \in \mathcal{X}/\sim$
Cohomological obstruction $H^1 \neq 0$	Projection degeneracy
Flat direction in $\mathcal{L}_{\text{local}}$	Regularizer flatness
Repair morphism trivializing c_{ij}	One step of the refinement loop
Sheaf repair operator $\mathcal{R}_{\text{sheaf}}$	Consistency operator $\mathcal{C}$
Stigmergic threshold	Contractivity condition on $\mathcal{C}$
Self-maintaining growth	Convergence of refinement loop to $[X^*]$

2.5 The Sheaf-Variational Equivalence

Definition 2.4 (Category of Projection Families). *Let $\mathbf{Proj}(\mathcal{X})$ be the category whose objects are pairs $(\mathcal{I}, \{\Pi_i\}_{i \in \mathcal{I}})$ and whose morphisms are inclusions of index sets compatible with the projection structure.*

Definition 2.5 (Category of Covering Sheaves). *Let $\mathbf{Sh}(\mathcal{M})$ be the category whose objects are sheaves $\mathcal{S}$ over covers $\mathcal{U}$ of $\mathcal{M}$, and whose morphisms are morphisms of sheaves over refinements of covers.*

Theorem 2.6 (Sheaf-Variational Equivalence). *There exists a functor*

$$\Psi : \mathbf{Proj}(\mathcal{X}) \longrightarrow \mathbf{Sh}(\mathcal{X}/\sim)$$

such that:

- (i) Ψ maps each projection family $\{\Pi_i\}_{i \in \mathcal{I}}$ to a sheaf $\mathcal{S}_{\Pi}$ over a cover of $\mathcal{X}/\sim$ whose local sections are observation-consistent field configurations.
- (ii) The gluing condition of $\mathcal{S}_{\Pi}$ is equivalent to the projection consistency condition $\Pi_i(X) = y_i$ for all i .
- (iii) The consistency operator $\mathcal{C}$ is the variational realization of the sheaf repair operator $\mathcal{R}_{\text{sheaf}}$ under Ψ .
- (iv) The fixed point $[X^*]$ of $\mathcal{C}$ corresponds under Ψ to the unique global section of $\mathcal{S}_{\Pi}$.

(v) The cohomological obstruction $H^1(\mathcal{U}, \mathcal{S}_\Pi) = 0$ if and only if the projection family is identifying on $\mathcal{F}$.

Proof. Construction of Ψ . Given a projection family $(\mathcal{I}, \{\Pi_i\})$, construct the cover $\mathcal{U} = \{U_i\}_{i \in \mathcal{I}}$ of $\mathcal{X}/\sim$ where

$$U_i = \{[X] \in \mathcal{A} \mid \|\Pi_i(X) - y_i\|_{\mathcal{Y}_i} \leq \epsilon_i\}.$$

Define the sheaf $\mathcal{S}_\Pi$ by assigning to each U_i :

$$\mathcal{S}_\Pi(U_i) = U_i \cap \mathcal{A}.$$

For inclusions $U_j \subseteq U_i$, the restriction map ρ_{ij} is the inclusion $U_j \cap \mathcal{A} \hookrightarrow U_i \cap \mathcal{A}$.

Part (i). By construction, $\mathcal{S}_\Pi(U_i)$ consists of admissible configurations whose i -th projection is within tolerance of y_i . ✓

Part (ii). A family $\{[X_i] \in \mathcal{S}_\Pi(U_i)\}$ satisfies the gluing condition iff $[X_i] = [X_j]$ on $U_i \cap U_j$ for all i, j , which is precisely the condition that a single $[X] \in \mathcal{A}$ satisfies $\Pi_i(X) = y_i$ for all i simultaneously. ✓

Part (iii). The objective of $\mathcal{R}_{\text{sheaf}}$ translates as:

$$\sum_{i < j} \mathcal{L}_{\text{seam}} + \mathcal{L}_{\text{local}} \longleftrightarrow \sum_i \|\Pi_i(X) - y_i\|^2 + \mathcal{R}(X),$$

which is exactly the objective of $\mathcal{C}$. ✓

Part (iv). A fixed point $[X^*]$ of $\mathcal{C}$ satisfies $[X^*] \in U_i$ for all i , giving a local section $s_i = [X^*]|_{U_i}$ that trivially satisfies the gluing condition. Uniqueness follows from Theorem 6.11. ✓

Part (v). $H^1(\mathcal{U}, \mathcal{S}_\Pi)$ measures failure of compatible local sections to glue into a global section, which is exactly projection degeneracy by part (ii). ✓

Functoriality. Adding sensors corresponds to refining the cover; the refinement map defines a natural transformation establishing functoriality. □ □

2.6 Consequences of the Equivalence

Consequence 1: The two failure modes are cohomological. Projection degeneracy is $H^1 \neq 0$; regularizer flatness is degenerate H^0 . Adding sensors can kill H^1 classes; strengthening the regularizer thins the stalks and kills degenerate H^0 .

Consequence 2: Dynamical disambiguation is cohomology in time. The dynamical admissibility set Γ extends the cover into the temporal direction. Trajectory separation kills residual H^1 classes that spatial sensors cannot. The full feasible set $\mathcal{F} = \Omega_{\text{obs}} \cap \Gamma$ is the global section of a sheaf over a spatio-temporal cover.

Consequence 3: Forward and inverse are dual. The forward direction studies the repair dynamics of $\mathcal{S}_\Pi$ from inside—how sections evolve under stigmergic reinforcement and homeorhetic flow. The inverse direction studies the existence and uniqueness of the global section from outside. Both study the same sheaf.

2.7 The Entropy Identification

The RSVP entropy field S appears in two roles throughout this framework: as the entropy component of the field triple $(\Phi, \mathbf{v}, S)$, and as a measure of unresolved obstruction. Under the Sheaf-Variational Equivalence these become one.

The observation-consistent set Ω_{obs} carries a natural measure induced by the Fisher information metric of the noise models $\{\mathcal{N}_i\}$ (developed in Appendix D). The volume of Ω_{obs} under this metric measures residual uncertainty. Under the equivalence:

$$\log \text{Vol}_{\text{Fisher}}(\Omega_{\text{obs}}) \longleftrightarrow \log \dim H^0(\mathcal{U}, \mathcal{S}_\Pi) \longleftrightarrow S_{\text{RSVP}}(X^*).$$

All three measure residual degrees of freedom after all constraints have been applied. Closure corresponds to each collapsing simultaneously. The RSVP entropy field is not merely a bookkeeping device—it is the local density of cohomological obstruction, and its minimization under $\mathcal{C}$ is the physical expression of sheaf repair.

3 The World-State Manifold and Its Symmetry Group

[PRELIMINARY OUTLINE — NOT YET DRAFTED]

This section develops the minimal mathematical infrastructure that Appendix B requires: the configuration space $\mathcal{X}$, its symmetry group G , the quotient $\mathcal{X}/\sim$, and the admissible set $\mathcal{A}$.

3.1 The Configuration Space. Defines $\mathcal{X} = H^1(\Omega) \times H^1(\Omega; \mathbb{R}^3) \times L^2(\Omega)$ over compact domain $\Omega \subset \mathbb{R}^3$. Specifies the product Sobolev norm. States that $\mathcal{X}$ is a separable Hilbert space and hence also a Fréchet manifold (this is elaborated in Appendix A). Identifies the three field components with physical quantities: Φ as ecosystem scalar density and civilizational legitimacy potential, $\mathbf{v}$ as resource flow and infrastructure transport vectors, S as entropy production and complexity budget. Gives explicit examples of each component in ecological and civilizational contexts to ground the abstraction before the symmetry group is introduced.

3.2 The Symmetry Group. Defines G as the group acting on $\mathcal{X}$ comprising: (a) gauge redundancies of the field representation—rephasing and rescaling transformations that leave physical observables invariant; (b) smooth coordinate reparametrizations of Ω —diffeomorphisms that relabel spatial positions without changing field content; (c) observationally equivalent reconfigurations of the civilizational layer—distinct infrastructure or demographic configurations that produce identical sensor outputs. States that the action of G is smooth, proper, and isometric with respect to the product Sobolev metric, so that $\mathcal{X}/G$ is Hausdorff and inherits a natural metric. Distinguishes G -equivalence from observational equivalence $\sim$: the former is intrinsic to the field representation, the latter is extrinsic and depends on the projection family. Notes that $\sim$ is coarser than G -equivalence in general and will collapse to G -equivalence when the projection family is complete (as established in Section 4).

3.3 The Admissible Set. Defines $\mathcal{A} \subset \mathcal{X}/\sim$ as the set of physically admissible equivalence classes: those satisfying the RSVP field equations, ecological plausibility constraints, and civilizational feasibility constraints. States Assumption A4 (geodesic convexity and strict convexity of $\mathcal{R}$ on $\mathcal{A}$) and gives the physical motivation for each condition: geodesic convexity ensures that admissible interpolations respect symmetry, lower semicontinuity ensures admissibility is stable under limits, coercivity ensures physically implausible configurations are excluded rather than merely penalized. Identifies $\mathcal{A}$ as a closed convex subset of the Banach realization $\mathcal{B}$ defined in Assumption B1 of Appendix B.

3.4 The RSVP Field Equations. States the plenum field equations governing time evolution of $(\Phi, \mathbf{v}, S)$: scalar continuity, vector momentum, and entropy production equations. These are the equations that $\mathcal{R}_{\text{RSVP}}$ penalizes violation of. Notes that solutions to these equations form the dynamical admissibility set Γ in

4 Sensor Modalities as Projection Operators

4.1 Purpose of This Section

Section 2 established that a projection family $\{\Pi_i\}_{i \in \mathcal{I}}$ and a sheaf $\mathcal{S}_\Pi$ over $\mathcal{X}/\sim$ are the same structure viewed from outside and inside respectively. What remains is to specify what the projection operators actually are—not as abstract bounded linear maps, but as concrete operators whose domains, codomains, and kernels are determined by the physics of each sensor modality.

This section defines each projection operator precisely enough that Assumption A1 can be verified directly. It establishes the RSVP correspondence: the identification of each modality with a specific component or functional of the field triple $(\Phi, \mathbf{v}, S)$. And it proves the completeness proposition: the condition under which the projection family collectively spans the full RSVP triple.

Nothing in this section is motivated by sensor engineering. The operators are defined as they are because those definitions are what the theorems in Sections 6 and 7 require.

4.2 The Configuration Space and Its Topology

Let $\Omega \subset \mathbb{R}^3$ be a compact domain with smooth boundary. The configuration space is

$$\mathcal{X} = H^1(\Omega) \times H^1(\Omega; \mathbb{R}^3) \times L^2(\Omega)$$

with product Sobolev norm $\|X\|_{\mathcal{X}}^2 = \|\Phi\|_{H^1}^2 + \|\mathbf{v}\|_{H^1}^2 + \|S\|_{L^2}^2$. All projection operators defined below are required to descend to the quotient $\mathcal{X}/\sim$: they must be constant on G -orbits.

4.3 Infrared Projection

Definition 4.1 (Infrared Projection). *The infrared projection $\Pi_{\text{IR}} : \mathcal{X} \rightarrow \mathcal{Y}_{\text{IR}}$ is defined by*

$$\Pi_{\text{IR}}(X) = (\Phi|_{\partial\Omega}, \nabla\Phi \cdot \hat{n}|_{\partial\Omega})$$

where $\hat{n}$ is the outward unit normal to $\partial\Omega$ and $\mathcal{Y}_{\text{IR}} = L^2(\partial\Omega) \times L^2(\partial\Omega)$.

Lemma 4.2. Π_{IR} is a bounded linear operator from $\mathcal{X}$ to $\mathcal{Y}_{\text{IR}}$.

Proof. The trace operator $\Phi \mapsto \Phi|_{\partial\Omega}$ is bounded from $H^1(\Omega)$ to $L^2(\partial\Omega)$ by the Sobolev trace theorem. The normal derivative is bounded from $H^1(\Omega)$ to $H^{-1/2}(\partial\Omega) \hookrightarrow L^2(\partial\Omega)$ for smooth $\partial\Omega$. $\square$

RSVP correspondence. Π_{IR} constrains boundary values and outward flux of Φ : energy density at the surface and rate of energy exchange. It does not directly constrain $\mathbf{v}$ or S in the interior.

4.4 LiDAR Projection

Definition 4.3 (LiDAR Projection). *Fix a threshold $\phi_0 > 0$. The LiDAR projection $\Pi_{\text{LiDAR}} : \mathcal{X} \rightarrow \mathcal{Y}_{\text{LiDAR}}$ is defined by*

$$\Pi_{\text{LiDAR}}(X) = \mathbf{1}_{\{\Phi \geq \phi_0\}} \quad \text{together with} \quad \Pi_{\text{LiDAR}}^\perp(X) = \left. \frac{\nabla \Phi}{|\nabla \Phi|} \right|_{\{\Phi = \phi_0\}}.$$

The observation space is $\mathcal{Y}_{\text{LiDAR}} = L^2(\Omega; \{0, 1\}) \times L^2(\{\Phi = \phi_0\}; S^2)$.

Lemma 4.4. Π_{LiDAR} is bounded for ϕ_0 a regular value of Φ .

Proof. For ϕ_0 a regular value, $\{\Phi = \phi_0\}$ is smooth by the implicit function theorem in H^1 . The indicator is bounded in $L^2(\Omega)$ by domain volume. The normal field is bounded by the Sobolev embedding $H^1 \hookrightarrow C^0$ in three dimensions. $\square$

RSVP correspondence. Π_{LiDAR} constrains the geometry of level sets of Φ and normal directions to those surfaces, which simultaneously constrain boundary conditions on $\mathbf{v}$: flows cannot cross material boundaries.

4.5 RGB/CCD Projection

Definition 4.5 (RGB Projection). *Let $\mathbf{c} \in S^2$ be the camera viewing direction and $f : \mathbb{R} \rightarrow \mathbb{R}^3$ a material reflectance function. The RGB projection $\Pi_{\text{RGB}} : \mathcal{X} \rightarrow \mathcal{Y}_{\text{RGB}}$ is defined by*

$$\Pi_{\text{RGB}}(X) = \int_0^\infty f(\Phi(\mathbf{x}(t))) e^{-\int_0^t \Phi(\mathbf{x}(s)) ds} dt$$

where $\mathbf{x}(t) = \mathbf{x}_0 + t\mathbf{c}$ and $\mathcal{Y}_{\text{RGB}} = L^2(\mathcal{V}; \mathbb{R}^3)$ over camera pixels $\mathcal{V}$.

Lemma 4.6. Π_{RGB} is bounded when f is Lipschitz and $\Phi \geq 0$.

Proof. The exponential attenuation is bounded by 1. The integrand is bounded by $\|f\|_\infty$, which is finite by Lipschitz continuity and $H^1 \hookrightarrow L^\infty$ in three dimensions. $\square$

RSVP correspondence. Π_{RGB} constrains Φ along rays—a projective complement to Π_{LiDAR} ’s level-set constraint. Their kernels overlap partially but not completely.

4.6 Audio Projection

Definition 4.7 (Audio Projection). *Let $\mathbf{x}_m \in \Omega$ for $m = 1, \dots, M$ be microphone positions. The audio projection $\Pi_{\text{audio}} : \mathcal{X} \times [0, T] \rightarrow \mathcal{Y}_{\text{audio}}$ is*

$$\Pi_{\text{audio}}(X)(t) = (\partial_t S(\mathbf{x}_m, t) + \nabla \cdot \mathbf{v}(\mathbf{x}_m, t))_{m=1}^M$$

with $\mathcal{Y}_{\text{audio}} = L^2([0, T]; \mathbb{R}^M)$.

RSVP correspondence. Π_{audio} is the only modality that directly constrains $\partial_t S$ —the rate of entropy production—and $\nabla \cdot \mathbf{v}$, encoding sources and sinks of flow. Hidden processes invisible to all spatial sensors are audible as entropy production events.

4.7 Photonic Strip Projection

Definition 4.8 (Photonic Strip Projection). *Let $\gamma : [0, L] \rightarrow \Omega$ parametrize the strip by arc length. The photonic strip projection $\Pi_{\text{strip}} : \mathcal{X} \times [0, T] \rightarrow \mathcal{Y}_{\text{strip}}$ is*

$$\Pi_{\text{strip}}(X)(s, t) = (\Phi(\gamma(s), t), \mathbf{v}(\gamma(s), t) \cdot \gamma'(s), \partial_t \Phi(\gamma(s), t))$$

with $\mathcal{Y}_{\text{strip}} = L^2([0, L] \times [0, T]; \mathbb{R}^3)$.

RSVP correspondence. The photonic strip provides continuous temporal records of Φ , tangential $\mathbf{v}$, and $\partial_t \Phi$ along a fixed transect: the only modality directly constraining temporal evolution of Φ at interior points.

4.8 The Completeness Proposition

Definition 4.9 (Spanning Condition). *A projection family $\{\Pi_i\}_{i \in \mathcal{I}}$ is field-spanning if*

$$\ker \tilde{\Pi} = \{[X] \in \mathcal{X}/\sim \mid \Pi_i(X) = 0 \forall i \in \mathcal{I}\} = \{[0]\}.$$

Proposition 4.10 (Completeness of the Five-Modality Family). *The projection family $\{\Pi_{\text{IR}}, \Pi_{\text{LiDAR}}, \Pi_{\text{RGB}}, \Pi_{\text{audio}}, \Pi_{\text{strip}}\}$ is field-spanning on $\mathcal{X}/\sim$.*

Proof. Suppose $[X] \in \ker \tilde{\Pi}$. $\Pi_{\text{IR}}(X) = 0$ gives $\Phi|_{\partial\Omega} = 0$. $\Pi_{\text{LiDAR}}(X) = 0$ gives $\Phi < \phi_0$ a.e. $\Pi_{\text{strip}}(X) = 0$ along strips dense in Ω gives $\Phi = 0$ on a dense subset; by H^1 continuity, $\Phi = 0$ on Ω . With $\Phi = 0$, strip vanishing also gives $\mathbf{v} \cdot \gamma' = 0$ along all strips; with three non-parallel strip directions, $\mathbf{v} = 0$. Then $\Pi_{\text{audio}}(X) = 0$ with $\mathbf{v} = 0$ gives $\partial_t S = 0$ at dense microphone positions, so S is time-constant; RSVP entropy constraints force $S = 0$. Hence $[X] = [0]$. $\square$

4.9 Degrees of Freedom Table

Modality	Field constrained	Where	New DOF closed
Infrared	$\Phi, \nabla\Phi \cdot \hat{n}$	$\partial\Omega$	Boundary energy, flux
LiDAR	Level sets of Φ , normals	Interior surfaces	Geometry; $\mathbf{v}$ BCs
RGB	Φ along rays	Lines of sight	Interior density
Audio	$\partial_t S, \nabla \cdot \mathbf{v}$	Point measurements	Temporal entropy, flow sources
Photonic strip	$\Phi, \mathbf{v} \cdot \gamma', \partial_t \Phi$	Transects over time	Temporal Φ ; tangential flow

4.10 Each Modality as a Chart

Under the Sheaf-Variational Equivalence, each Π_i defines a chart $U_i = \{[X] \in \mathcal{A} \mid \|\Pi_i(X) - y_i\| \leq \epsilon_i\}$ on $\mathcal{X}/\sim$. The five modalities define five charts. Completeness ensures the cover is fine enough to support a non-degenerate sheaf. Sensor design is sheaf design: choosing modalities is choosing the cover of $\mathcal{X}/\sim$.

5 The Feasible Set and Its Geometry

[PRELIMINARY OUTLINE — NOT YET DRAFTED]

This section defines and analyzes the geometric object whose dimensional collapse is the operative notion of closure throughout the paper.

4.1 The Observation-Consistent Set. Defines $\Omega_{\text{obs}} = \{[X] \in \mathcal{A} \mid \|\Pi_i(X) - y_i\|_{\mathcal{Y}_i} \leq \epsilon_i \forall i\}$ and proves it is non-empty (Lemma 5.1), closed, and convex (Lemma 5.2), using results already established in Section 6. Gives geometric intuition: Ω_{obs} is the intersection of five “tubes” in $\mathcal{X}/\sim$, one per sensor modality, each of radius ϵ_i around the preimage of y_i under Π_i .

4.2 The Dynamical Admissibility Set. Introduces Γ as a named object whose full definition appears in Section 7. States its key property here: Γ is a subset of $\mathcal{A}$ consisting of configurations that lie on admissible trajectories of the world dynamics. Notes that $\Gamma \subseteq \mathcal{A}$ by forward-invariance (Lemma 6.1), and that in the deterministic limit Γ^0 is the union of integral curves of the vector field F .

4.3 The Feasible Set. Defines $\mathcal{F} = \Omega_{\text{obs}} \cap \Gamma$ as the primary geometric object of the paper. Notes that $\mathcal{F}$ is the correct domain for both theorems: the Identifiability Theorem requires $\tilde{\Pi}$ to be injective on $\mathcal{F}$, not on all of $\mathcal{X}/\sim$; the Disambiguation Theorem requires trajectory separation on $\mathcal{F}$, not on Ω_{obs} alone.

4.4 Topology of the Feasible Set. Analyzes the topology of $\mathcal{F}$ as a function of the projection family and the ϵ_i profile. The key quantity is $\dim \mathcal{F}$: closure corresponds to $\dim \mathcal{F} \rightarrow 0$. States and proves the Feasible Set Contraction theorem: under monotone addition of sensors with independent noise models, $\dim \mathcal{F}$ is non-increasing, with strict decrease when the new sensor constrains at least one degree of freedom not already closed by $\{\Pi_1, \dots, \Pi_{i-1}\}$. Connects this to the completeness result of Section 4: the five-modality family is complete iff it can in principle drive $\dim \mathcal{F}$ to zero.

4.5 Information Geometry of the Feasible Set. Introduces the Fisher information metric on $\mathcal{F}$ induced by the noise models $\{\mathcal{N}_i\}$. Under this metric, $\mathcal{F}$ is a Riemannian manifold and its volume $\text{Vol}_{\text{Fisher}}(\mathcal{F})$ is the measure of residual uncertainty. States that closure corresponds to $\text{Vol}_{\text{Fisher}}(\mathcal{F}) \rightarrow 0$. Identifies this volume with the RSVP entropy and cohomological obstruction as established in Section 2. Full development of the Fisher geometry is deferred to Appendix D.

4.6 The Sheaf Interpretation. Under the Sheaf-Variational Equivalence, $\mathcal{F}$ is the intersection of stalks across all charts of the cover $\mathcal{U}$. Closure $|\mathcal{F}| = 1$ is exactly the sheaf condition: there exists a unique global section consistent across all charts. Non-existence of a global section is not a failure mode of the system—it is a falsification signal indicating that the model class $\mathcal{A}$ needs revision. This is an instance of the general principle from sheaf theory: non-trivial H^1 is informative, not merely obstructive.

6 The Consistency Operator and the Identifiability Theorem

6.1 Setup and Standing Assumptions

Let $\mathcal{X} = H^1(\Omega) \times H^1(\Omega; \mathbb{R}^3) \times L^2(\Omega)$ with the product Sobolev topology (Appendix A). A symmetry group G acts smoothly, properly, and isometrically on $\mathcal{X}$; the quotient $\mathcal{X}/\sim$ is Hausdorff. We work in $\mathcal{X}/\sim$ throughout.

Assumption 6.1 (Projection Regularity). *Each $\Pi_i : \mathcal{X} \rightarrow \mathcal{Y}_i$ is a bounded linear map to a separable Hilbert space $\mathcal{Y}_i$.*

Assumption 6.2 (Noise Model). *Each observation satisfies $y_i = \Pi_i(X^*) + \eta_i$ where $\|\eta_i\|_{\mathcal{Y}_i} \leq \epsilon_i$ almost surely.*

Assumption 6.3 (Regularizer Structure). *The regularizer takes the form*

$$\mathcal{R}(X) = \mathcal{R}_{\text{RSVP}}(X) + \lambda_{\text{eco}} \mathcal{R}_{\text{eco}}(X) + \lambda_{\text{civ}} \mathcal{R}_{\text{civ}}(X)$$

where $\mathcal{R}_{\text{RSVP}}$ measures violation of the RSVP field equations in L^2 , $\mathcal{R}_{\text{eco}}$ is zero on ecologically admissible configurations and strictly positive off them, and $\mathcal{R}_{\text{civ}}$ is zero on civilizationaly admissible configurations and strictly positive off them.

Assumption 6.4 (Admissibility Region). *The admissible set $\mathcal{A} = \{[X] \in \mathcal{X}/\sim \mid \mathcal{R}(X) < \infty\}$ is a closed convex subset of $\mathcal{X}/\sim$, and $\mathcal{R}$ restricted to $\mathcal{A}$ is strictly convex along geodesics.*

Assumption 6.5 (Banach Realization). *There exists a Banach space $\mathcal{B}$ and continuous embedding $\iota : \mathcal{X} \hookrightarrow \mathcal{B}$ such that $\mathcal{A}$ is closed and convex in $\mathcal{B}$, each Π_i extends to a bounded linear operator on $\mathcal{B}$, and $\mathcal{R}$ extends to a proper lower semicontinuous functional on $\mathcal{B}$.*

In practice one takes $\mathcal{B} = \mathcal{X}$ itself, which is already a Hilbert space. The Banach realization is stated separately to make the compactness and weak convergence arguments in Appendix B self-contained.

6.2 The Feasible Set

Definition 6.6 (Observation-Consistent Set).

$$\Omega_{\text{obs}} = \{[X] \in \mathcal{A} \mid \|\Pi_i(X) - y_i\|_{\mathcal{Y}_i} \leq \epsilon_i \ \forall i \in \mathcal{I}\}.$$

Lemma 6.7 (Non-emptiness). *Under Assumptions 6.1 and 6.2, $X^* \in \Omega_{\text{obs}}$ almost surely.*

Proof. $\|y_i - \Pi_i(X^*)\| = \|\eta_i\| \leq \epsilon_i$ almost surely by Assumption 6.2. $\square$

Lemma 6.8 (Closedness and Convexity). *Under Assumptions 6.1 and 6.4, Ω_{obs} is a closed convex subset of $\mathcal{A}$.*

Proof. Each constraint set is closed (continuity of Π_i) and convex (linearity of Π_i , convexity of norm ball). Intersection of closed convex sets is closed and convex. $\square$

The dynamically admissible set Γ is introduced as a named object here; its full definition and construction are given in Section 7. The *feasible set* is $\mathcal{F} = \Omega_{\text{obs}} \cap \Gamma$.

6.3 The Consistency Operator

Definition 6.9 (Consistency Operator).

$$\mathcal{C}(X) = \arg \min_{[X'] \in \mathcal{A}} \left[\sum_{i \in \mathcal{I}} \|\Pi_i(X') - y_i\|_{\mathcal{Y}_i}^2 + \mathcal{R}(X') \right].$$

$\mathcal{C}$ maps $\mathcal{A}$ to $\mathcal{A}$: the optimization is constrained to the admissible region. A configuration minimizing projection error but violating RSVP equations or ecological plausibility is not a candidate world-state. The regularizer defines the domain of the operator, not a preference over an unconstrained space.

Lemma 6.10 (Well-definedness). *Under Assumptions 6.1–6.5, $\mathcal{C}(X)$ exists and is unique for every $[X] \in \mathcal{A}$.*

Proof. The objective $f(X') = \sum_i \|\Pi_i(X') - y_i\|^2 + \mathcal{R}(X')$ is continuous, strictly convex (sum of convex functions plus strictly convex $\mathcal{R}$), and coercive on $\mathcal{A}$. A continuous strictly convex coercive function on a closed convex set attains its minimum at a unique point. Full proof in Appendix B. $\square$

6.4 The Identifiability Theorem

The induced map of the projection family on the quotient is:

$$\tilde{\Pi} : \mathcal{X}/\sim \longrightarrow \prod_{i \in \mathcal{I}} \mathcal{Y}_i, \quad [X] \mapsto (\Pi_i(X))_{i \in \mathcal{I}}.$$

By definition of $\sim$, $\tilde{\Pi}$ is well-defined.

Theorem 6.11 (Yarncrawler Identifiability). *Under Assumptions 6.1–6.5, the consistency operator $\mathcal{C}$ has a unique fixed point $[X^*] \in \mathcal{F}$ if and only if $\tilde{\Pi}$ is injective on $\mathcal{F}$.*

Proof. ($\Rightarrow$) Suppose $\mathcal{C}$ has a unique fixed point $[X^*] \in \mathcal{F}$. If $\tilde{\Pi}$ were not injective on $\mathcal{F}$, there exist distinct $[X], [X'] \in \mathcal{F}$ with $\Pi_i(X) = \Pi_i(X')$ for all i . Both achieve zero projection residual in the objective. The objective reduces to $\mathcal{R}(X)$ vs. $\mathcal{R}(X')$. Either $\mathcal{R}(X) = \mathcal{R}(X')$ —violating strict convexity of $\mathcal{R}$ on $\mathcal{A}$ (Assumption 6.4)—or one has strictly smaller regularizer value, making the other not a minimizer. Either way contradicts uniqueness.

($\Leftarrow$) If $\tilde{\Pi}$ is injective on $\mathcal{F}$, then $\Omega_{\text{obs}} \cap \mathcal{F}$ contains at most one point. By Lemma 5.1, $[X^*] \in \Omega_{\text{obs}}$; by definition $[X^*] \in \Gamma$; hence $[X^*] \in \mathcal{F}$. Injectivity forces $|\mathcal{F}| = 1$. At $[X^*]$ all projection residuals are within tolerance and $\mathcal{R}(X^*)$ is minimized over the singleton $\mathcal{F}$. Hence $\mathcal{C}([X^*]) = [X^*]$. $\square$

6.5 The Corollary on Diagnosable Failure Modes

Corollary 6.12. *Identifiability fails in exactly two geometrically distinguishable ways:*

(i) Projection degeneracy: $\tilde{\Pi}$ is not injective on $\mathcal{F}$. Two distinct admissible world-states produce identical observations. This failure is intrinsic to the sensor family and cannot be resolved by the regularizer.

(ii) Regularizer flatness: $\mathcal{R}$ is not strictly convex on $\mathcal{F}$. The regularizer assigns equal admissibility to configurations the sensors cannot distinguish. This failure is intrinsic to the model class.

These failure modes are geometrically distinguishable: projection degeneracy corresponds to a non-trivial kernel of $\tilde{\Pi}|_{\mathcal{F}}$; regularizer flatness corresponds to a flat direction of $\mathcal{R}$ within $\mathcal{F}$. They can coexist.

6.6 Why the Quotient Structure Is Not Optional

In a standard inverse problem one works in $\mathcal{X}$ directly. But in the Yarncrawler setting, two configurations related by a gauge transformation or coordinate reparametrization represent the same physical situation. Working in $\mathcal{X}$ rather than $\mathcal{X}/\sim$ would make uniqueness false by construction—there would always be a continuum of configurations achieving the same objective, related by the action of G .

An L^2 uniqueness statement conflates field-theoretic gauge redundancy with measurement noise. A measure-theoretic statement collapses the geometry of $\mathcal{X}/\sim$ into observational statistics and loses the regularizer’s role as an admissibility condition. The symmetry-group formulation is not a choice among equivalent formalisms; it is the only formulation that preserves the semantic commitments of the framework.

7 Trajectory Learning and the Dynamical Admissibility Set

7.1 The Central Obstruction

Section 6 established that $\mathcal{C}$ has a unique fixed point iff $\tilde{\Pi}$ is injective on $\mathcal{F}$. This condition is clean but generically unmet. Under realistic sensing, Ω_{obs} is high-dimensional. Projection degeneracy is the default condition, not a pathological case.

The resolution cannot come from within the static framework. What is required is a constraint of a different kind—one operating not on the configuration X at a single time but on the trajectory through which X was reached. The dynamical admissibility set Γ is not an additional observation; it is a restriction on which configurations are reachable under the dynamics of the world. Two configurations in Ω_{obs} that are observationally indistinguishable at one moment may evolve under those dynamics into configurations that are distinguishable at a later moment.

7.2 The Path Space and the Stochastic Kernel

Definition 7.1 (Discrete Path Space). $\mathcal{P}_\Delta = \{\mathbf{X} = ([X_0], [X_1], \dots, [X_N]) \mid [X_k] \in \mathcal{A}, N = \lfloor T/\Delta \rfloor\}$.

Definition 7.2 (Stochastic Transition Kernel). $T_\Delta : \mathcal{A} \rightarrow \mathcal{P}(\mathcal{A})$, where for each $[X] \in \mathcal{A}$, $T_\Delta([X], \cdot)$ specifies the distribution over one-step successors under the world dynamics at resolution Δ .

Definition 7.3 (Dynamically Admissible Set).

$$\Gamma = \{[X] \in \mathcal{A} \mid \exists [X_0] \in \mathcal{A}, \exists k \geq 0 \text{ s.t. } \mu_{X_0}^{T_\Delta}([X_k] = [X]) > 0\}.$$

Γ is the support of the union of all admissible path measures. It is a geometric object—a subset of $\mathcal{A}$ —not a probabilistic one.

Lemma 7.4 (Invariance of Γ). If T_Δ is time-homogeneous and $\mathcal{A}$ is forward-invariant under T_Δ , then $\Gamma \subseteq \mathcal{A}$ and Γ is forward-invariant.

7.3 The Deterministic Limit and Tangent Bundle Constraint

Assumption 7.5 (Generator Regularity). There exists a Lipschitz vector field $F(\cdot; \theta) : \mathcal{A} \rightarrow T\mathcal{A}$ such that $T_\Delta([X], \cdot) \Rightarrow \delta_{[X] + \Delta F([X]; \theta)}$ weakly as $\Delta \rightarrow 0$.

In the zero-noise limit, the dynamically admissible set reduces to:

$$\Gamma^0 = \{[X] \in \mathcal{A} \mid \exists [X_0] \in \mathcal{A}, \exists t \in [0, T] \text{ s.t. } \Phi_F^t([X_0]) = [X]\}$$

where Φ_F^t is the flow of F at time t .

Definition 7.6 (Admissible Tangent Cone). *For $[X] \in \Gamma^0$: $\mathcal{T}_{[X]}\Gamma^0 = \{v \in T_{[X]}\mathcal{A} \mid v = F([X]; \theta) \text{ for some } \theta \in \Theta\}$.*

Proposition 7.7 (Tangent Dimension Reduction). *If $\Theta \subset \mathbb{R}^p$, then $\dim \mathcal{T}_{[X]}\Gamma^0 \leq p$. In particular, if $p < \dim T_{[X]}\mathcal{A}$, the dynamical constraint reduces the effective dimensionality of admissible directions from $\dim T_{[X]}\mathcal{A}$ to at most p .*

Proof. The map $\theta \mapsto F([X]; \theta)$ is smooth from $\Theta \subset \mathbb{R}^p$ to $T_{[X]}\mathcal{A}$; its image has dimension at most p by the rank theorem. $\square$

7.4 Trajectory Learning from Before/After Pairs

Definition 7.8 (Fragment-Consistent Parameter Set). *Given fragment $(y_{\text{before}}, y_{\text{after}}, \tau)$:*

$$\Theta_{(y_b, y_a, \tau)} = \{\theta \in \Theta \mid \exists [X_b] \in \Omega_{\text{obs}}(y_b), \Pi_i(\Phi_F^\tau([X_b])) \in B_{\epsilon_i}(y_{a,i}) \forall i\}.$$

Given training set $\mathcal{D}$, the learned parameter set is $\hat{\Theta} = \bigcap_n \Theta_{(y_b^{(n)}, y_a^{(n)}, \tau^{(n)})}$.

The operator class library. In practice $F(\cdot; \theta)$ is decomposed as:

$$F([X]; \theta) = \sum_{k=1}^K \alpha_k([X]) \cdot F_k([X]; \theta_k)$$

where each F_k corresponds to a named ecological or civilizational process—forest succession, urban expansion, agricultural cycling, infrastructure decay, hydrological change, demographic transition. Each F_k is learned from before/after pairs in which that process is dominant. The activation weights $\alpha_k([X]) \geq 0$ satisfy $\sum_k \alpha_k = 1$.

7.5 The Dynamical Disambiguation Theorem

Definition 7.9 (Trajectory Separation). *Two configurations $[X], [X'] \in \Omega_{\text{obs}}$ are trajectory-separated by $F(\cdot; \hat{\theta})$ at time $\tau > 0$ if*

$$\left\| \tilde{\Pi}(\Phi_F^\tau([X]; \hat{\theta})) - \tilde{\Pi}(\Phi_F^\tau([X']; \hat{\theta})) \right\|_{\prod_i \mathcal{Y}_i} > \epsilon$$

for some $\epsilon > 0$ depending only on the noise levels $\{\epsilon_i\}$.

Theorem 7.10 (Dynamical Disambiguation). *Under Assumptions 6.1–7.5, if $|\Omega_{\text{obs}}| > 1$ and every pair $[X], [X'] \in \Omega_{\text{obs}}$ with $[X] \neq [X']$ is trajectory-separated by $F(\cdot; \hat{\theta})$ at some $\tau \in (0, T]$, then $|\mathcal{F}| = |\Omega_{\text{obs}} \cap \Gamma^0| = 1$.*

Proof. Let $[X], [X'] \in \Omega_{\text{obs}} \cap \Gamma^0$ be distinct. Since both lie in Γ^0 , both lie on integral curves of $F(\cdot; \hat{\theta})$. By trajectory separation, there exists $\tau \in (0, T]$ such that the forward-evolved projections differ by more than ϵ . At time τ , observed data $y(\tau)$ must lie within ϵ_i of the true trajectory; therefore at most one of $[X(\tau)], [X'(\tau)]$ lies in $\Omega_{\text{obs}}(y(\tau))$. The true state $[X^*(\tau)] \in \Omega_{\text{obs}}(y(\tau))$ by Lemma 5.1 at time τ . Applying to all pairs eliminates all but $[X^*]$. Hence $|\mathcal{F}| = 1$. $\square$

7.6 Joint Identifiability

Corollary 7.11 (Joint Identifiability). *The consistency operator $\mathcal{C}$ restricted to $\mathcal{F}$ has a unique fixed point if either: (i) $\tilde{\Pi}$ is injective on $\mathcal{F}$; (ii) every pair in $\Omega_{\text{obs}} \cap \Gamma^0$ is trajectory-separated; or (iii) both conditions hold partially. The sufficient condition for (iii) is that $\dim(\Omega_{\text{obs}} / \ker \tilde{\Pi}) + \dim(\Gamma^0 / \Gamma_{\text{indist}}^0) \geq \dim \Omega_{\text{obs}}$.*

Case (iii) is the generic realistic scenario: sensors eliminate most ambiguity, trajectory learning eliminates the residual. The two mechanisms are additive in their dimensional reduction, not competing.

8 The Inversion Layer

[PRELIMINARY OUTLINE — NOT YET DRAFTED]

This section develops the inversion layer as an architecture for approximating the consistency operator in practice, without conflating the approximation with the operator itself.

7.1 Role Specification. States the division of labor precisely: the inversion layer is not the model, it is a proposal mechanism. Its role is to place the iterative optimization in the basin of the correct fixed point. The consistency operator $\mathcal{C}$ is the model. The encoder is a high-entropy initializer; the physics-constrained forward model is the contraction operator. Explains why conflating encoder with solution is the failure mode of standard encoder-decoder stacks: doing so makes the failure modes of Corollary 6.12 undiagnosable. Separating proposal from decision preserves the falsifiability structure.

7.2 The Encoder as Approximate Posterior. Defines the encoder as implementing an approximate posterior:

$$q_\phi(X \mid \{y_i\}) \approx p(X \mid \{y_i\}) \propto \exp\left(-\sum_i d_i(\Pi_i(X), y_i) - \mathcal{R}(X)\right).$$

The encoder approximates this distribution, not solves it. The forward-consistency loop is what contracts that distribution. Discusses the variational inference interpretation: the encoder minimizes KL divergence from the true posterior, subject to the architectural constraints of the learned network. Notes that because $\mathcal{R}$ encodes physical admissibility, the approximate posterior is already biased toward physically plausible configurations—the encoder does not need to learn physics from scratch.

7.3 The Refinement Loop. Specifies the five-step loop:

1. Encoder samples $X^{(0)} \sim q_\phi(\cdot \mid \{y_i\})$
2. Forward model computes $\{\hat{y}_i\} = \{\Pi_i(X^{(0)})\}$
3. Residuals $\{r_i\} = \{d_i(\hat{y}_i, y_i)\}$ drive gradient update
4. $X^{(k+1)} = \mathcal{C}(X^{(k)})$ (one step of projected gradient descent on f)
5. Iterate until $\dim \mathcal{F} < \varepsilon$ (closure criterion from Section 9)

Discusses convergence: Theorem 6.11 guarantees a unique fixed point when conditions hold; the refinement loop approximates the path to that fixed point. Convergence rate depends on the condition number of the Hessian of f at $[X^*]$, which is related to the Fisher information geometry of Appendix D.

7.4 The Categorical Structure of ²⁴Multi-Modal Fusion. Develops the secondary, structural description: the encoder as a collection of modality-specific proposal maps composed via a lax monoidal structure over $\Pi \mathcal{M}$. Each modality

9 Closure: Definition, Criteria, and Detection

[PRELIMINARY OUTLINE — NOT YET DRAFTED]

This section defines closure operationally and proves that the three closure criteria are equivalent under the main theorems.

8.1 The Unified Definition. A state $[X^*] \in \mathcal{X}/\sim$ is *closed* if it is the unique fixed point of $\mathcal{C}$ and lies on a trajectory of T_Δ consistent with all observed fragments. This definition is then decomposed into three conditions—projection parity, dynamical stability, counterfactual consistency—which are the operational handles on closure in a deployed system.

8.2 The Three Conditions. *Projection parity:* $\|\Pi_i(X^*) - y_i\|_{\mathcal{Y}_i} \leq \epsilon_i$ for all $i \in \mathcal{I}$. This is the basic requirement that the fixed point is consistent with all current observations. It is verifiable in real time.

Dynamical stability: $[X^*] \in \Gamma^0$, meaning $[X^*]$ lies on an integral curve of $F(\cdot; \hat{\theta})$. Equivalently, the fixed point is not merely consistent with instantaneous observations but is reachable under the learned dynamics. This rules out configurations that happen to fit current observations but could not have been produced by the world’s actual trajectory.

Counterfactual consistency: the system must be stable under re-projection. If $[X^*]$ is simulated forward under F for time τ and the resulting $[\hat{X}^*(\tau)]$ is projected back to observation space, the result must lie within ϵ_i of the actual observations at time τ . This is the strongest condition: it requires not just that the fixed point is consistent with past observations but that its forward trajectory is consistent with future observations.

8.3 Equivalence Theorem. States and proves that the three conditions are equivalent under the assumptions of Theorems 6.11 and 7.10. The key step is showing that each condition implies $\dim \mathcal{F} = 0$, and that $\dim \mathcal{F} = 0$ implies all three. The proof uses the identification from Section 2: the three conditions are projection parity (Ω_{obs} tight), dynamical stability (Γ non-trivially intersecting Ω_{obs}), and counterfactual consistency (the intersection is trajectory-stable, not just pointwise). These are three faces of $|\mathcal{F}| = 1$.

8.4 Operational Detection. Closure is operationally detectable as stability of $[X^*]$ under perturbation of ϵ_i : if tightening any constraint by $\delta\epsilon_i$ does not move the fixed point, closure has been achieved. This gives a practical stopping criterion for the refinement loop of Section 8. Discusses the rate of convergence to closure as a function of sensor density and trajectory fragment coverage, connecting to the Fisher information geometry of Appendix D.

8.5 Closure and Entropy Collapse. Connects the closure detection criterion to the entropy identification of Section 2: at closure, $\text{Vol}_{\text{Fisher}}(\mathcal{F}) = 0$, $H^1(\mathcal{U}, \mathcal{S}_\Pi) = 0$, $\dim H^0 = 1$, and $S_{\text{RSVP}}(X^*)$ is minimized subject to admissibility—simultaneously and equivalently. The closed state is not a best estimate; it is the unique trajectory

10 Deployment Scales as Feasible Set Geometry

[PRELIMINARY OUTLINE — NOT YET DRAFTED]

This section unifies planetary, regional, and site-scale deployment as the same system with different geometries of the feasible set.

9.1 The Unified Framework. States the central claim: all three deployment scales are instances of the same consistency operator $\mathcal{C}$ with different $(\Pi_i, \mathcal{N}_i, \epsilon_i)$ profiles. The scale is not a separate architectural choice—it is a parameter of the noise model. A planetary-scale deployment has large ϵ_i (sparse, low-frequency observations); a site-scale deployment has small ϵ_i (dense, high-frequency observations). The feasible set Ω_{obs} has different geometry at each scale, but the identifiability theorem and disambiguation theorem apply uniformly.

9.2 Planetary Scale. Sensors: satellite multispectral imagery (RGB and infrared), occasional LiDAR passes, no audio, no photonic strips. Temporal resolution: weeks to months. Spatial resolution: 10–100 meters per pixel. Noise levels ϵ_i are large. Consequence: Ω_{obs} is high-dimensional and weakly constrained. The system operates in the high-entropy proposal regime, relying heavily on $\mathcal{R}$ to restrict $\mathcal{F}$. Trajectory learning carries most of the disambiguation burden: seasonal cycles, annual land cover change, and multi-year succession patterns provide the operator class library entries that contract Γ . States the modified identifiability condition for this regime: the projection family need not be complete (Proposition 4.10 fails at planetary scale due to missing audio and strip modalities), but trajectory learning can compensate via Corollary 6.4(iii).

9.3 Regional Scale. Sensors: aerial multispectral and LiDAR surveys, possibly distributed acoustic sensors, periodic photonic strip transects. Temporal resolution: days to weeks. Spatial resolution: 1–10 meters. Noise levels intermediate. Consequence: Ω_{obs} contracts along spatial dimensions but remains temporally undersampled. The system requires both static projection richness (more modalities available than at planetary scale) and trajectory learning. The operator class library is expanded: finer-grained process operators for agricultural change, urban growth, forest disturbance. Discusses the transition from planetary to regional scale as a cover refinement: the planetary-scale $\mathcal{S}_{\Pi}$ restricts to the regional-scale $\mathcal{S}_{\Pi'}$ on the finer cover, with H^1 classes killed by the additional modalities.

9.4 Site Scale. Sensors: fixed arrays including all five modalities at high spatial and temporal density. Temporal resolution: seconds to minutes. Spatial resolution: centimeters to meters. Noise levels ϵ_i small. Consequence: Ω_{obs} collapses rapidly toward a singleton. The system achieves genuine closure and can generate counterfactual re-projections for real-time validation. Discusses the calibration loop: at site scale, re-projection residuals (forward model outputs vs. actual sensor readings) provide a continuous closure detection signal, implementing the operational criterion of Section 9.

11 Relation to Prior Work

[PRELIMINARY OUTLINE — NOT YET DRAFTED]

This section positions Yarncrawler against four traditions: data assimilation, inverse rendering, digital twin systems, and categorical/sheaf-theoretic semantics. The goal is not to survey those traditions but to identify precisely where Yarncrawler departs from each and why the departure is necessary.

10.1 Data Assimilation and Ensemble Methods. Data assimilation—especially ensemble Kalman filtering (EnKF) and variational methods (4D-Var)—solves the same broad problem: estimating a hidden state from observations over time. The key differences are: (a) data assimilation works in $\mathcal{X}$ rather than $\mathcal{X}/\sim$, treating gauge redundancies as noise rather than structure; (b) the regularizer in data assimilation is a prior distribution, not an admissibility condition—it expresses preference, not exclusion; (c) data assimilation has no sheaf structure: there is no notion of local charts, restriction maps, or cohomological obstruction. The Yarncrawler Identifiability Theorem has no counterpart in standard data assimilation because the quotient structure is absent. Notes where EnKF and 4D-Var are special cases: in the linear Gaussian regime with trivial symmetry group ($G = \{e\}$), $\mathcal{C}$ reduces to the Kalman update.

10.2 Inverse Rendering and Neural Radiance Fields. Neural radiance fields (NeRF) and related inverse rendering methods solve the problem of reconstructing a 3D scene from 2D images. The RGB projection operator Π_{RGB} defined in Section 4 is the NeRF rendering integral. The differences are: (a) NeRF has no dynamics—it is a purely static reconstruction method with no Γ ; (b) NeRF has no RSVP field structure—the scene representation is a learned radiance function, not a physical field triple; (c) NeRF has no regularizer encoding physical admissibility. The Yarncrawler framework strictly generalizes NeRF: the RGB projection is one chart of the cover, and NeRF is the degenerate case where only that chart is available and dynamics are absent.

10.3 Digital Twin Systems. Digital twin literature proposes maintaining synchronized virtual replicas of physical systems. The standard approach is object-ontological: the twin represents a collection of objects (buildings, trees, vehicles) whose properties are updated from sensor data. Yarncrawler departs in three ways: (a) the primary ontology is trajectory-based, not object-based—objects emerge as persistent patterns in field trajectories, not as primary entities; (b) the twin is not updated by recognized objects but by the consistency operator acting on the full quotient configuration space; (c) digital twins have no identifiability theory—there is no formal account of when a twin uniquely determines its physical counterpart. The Yarncrawler Identifiability Theorem²⁷ provides exactly this.

10.4 Categorical and Sheaf-Theoretic Semantics. Sheaf theory has been

12 Discussion: Closure as Epistemology

[PRELIMINARY OUTLINE — NOT YET DRAFTED]

This section draws the epistemological consequence of the framework and connects it to the broader Yarncrawler theoretical program.

11.1 What Closure Means. A closed Yarncrawler state is not a probabilistic estimate. It is not the most likely world-state given the observations. It is the unique world-state that is structurally demanded by the observations under the model class. The distinction is not merely philosophical: it has operational consequences. A probabilistic estimate can always be revised by new observations; a closed state cannot be revised without either adding new sensor modalities (killing new H^1 classes) or revising the model class (changing $\mathcal{A}$). The closed state is, in a precise sense, as certain as the model class allows.

11.2 Constraint Primacy. The framework operationalizes a philosophical commitment: constraints are prior to completions. The world-state is not recovered by filling in a model template with observations—it is the unique configuration that survives after all constraints have been applied. This is the physical analogue of a principle that recurs throughout the Yarncrawler program: meaning emerges from constraint closure rather than from representation. The Sheaf-Variational Equivalence makes this precise: a world-state has meaning (is uniquely identified) exactly when the sheaf induced by its projections has a unique global section.

11.3 The RSVP Fields as Constraint Carriers. The RSVP field triple $(\Phi, \mathbf{v}, S)$ is not merely a convenient parametrization of the configuration space. It is the structure that makes the regularizer $\mathcal{R}$ non-arbitrary: the admissible set $\mathcal{A}$ is defined by the RSVP field equations, which encode physical conservation laws, ecological succession principles, and civilizational feasibility constraints. Without the RSVP structure, the regularizer would be an arbitrary penalty function and the closed state would be an artifact of the choice of penalty. With it, the closed state is the unique configuration consistent with the laws governing the domain. The framework is therefore not just an inference procedure—it is a theory of what it means for a physical configuration to be real.

11.4 Stigmergic Accumulation and the Refinement Loop. Connects the refinement loop of Section 8 to the stigmergic dynamics developed in the Yarncrawler forward direction. Each step of the refinement loop is a repair morphism in the sheaf $\mathcal{S}_{\Pi}$: it reduces a projection residual (trivializes a cocycle) and updates the field configuration toward closure. The accumulation of constraints over time—as more sensor observations arrive—is the variational analogue of stigmergic reinforcement: each new observation narrows the feasible set, and the trajectory of the refinement loop through configuration space is the path the crawler takes through the yarn. The loop converges when no more constraints can be satisfied—when the yarn has been fully crawled and the global section is unique.

13 Conclusion

[PRELIMINARY OUTLINE — NOT YET DRAFTED]

The conclusion is short—three to four paragraphs. It does not summarize the paper. It states what the paper has established, what remains to be done, and what the framework implies.

Paragraph 1: What has been established. The Yarncrawler Identifiability Theorem and the Dynamical Disambiguation Theorem together characterize the conditions under which multi-modal sensor observations uniquely determine a physical world-state. The Sheaf-Variational Equivalence shows that this characterization is not merely a technical result about inverse problems: it is the variational realization of a categorical structure in which sensor modalities are charts, observations are local sections, and closure is the existence of a unique global section. The RSVP field triple is the substrate in which these structures are grounded.

Paragraph 2: The main finding compressed. A world-state is uniquely recoverable from multi-modal observations if and only if the induced map from the quotient configuration space to the product of observation spaces is injective on the feasible set, or if the dynamics of the world separate configurations that static sensors cannot. In either case, closure corresponds to cohomology trivialization, entropy collapse, and Fisher volume collapse—simultaneously and equivalently.

Paragraph 3: What remains. Three open problems are identified. (a) The extension of the identifiability theorem to the stochastic case without the low-noise limit: does a version of the theorem hold when σ^2 is not small? (b) The learning theory of $F(\cdot; \hat{\theta})$: what sample complexity is needed for the operator class library to ensure trajectory separation with high probability? (c) The connection between the Fisher information geometry of the feasible set and the computational complexity of the refinement loop: can the convergence rate of the refinement loop be bounded in terms of the eigenspectrum of the Fisher metric at $[X^*]$?

Paragraph 4: The implication. The Yarncrawler system is not a sensing architecture. It is a theory of when the world becomes legible—when the observations accumulated by a sensor network, constrained by physical law and dynamical history, leave no alternative configuration standing. That condition is not an approximation target. It is a threshold: on one side, multiple histories remain consistent with the evidence; on the other, only one does. The framework identifies that threshold precisely and provides the operator that reaches it.

A Fréchet Manifold Structure of $\mathcal{X}$

[PRELIMINARY OUTLINE — NOT YET DRAFTED]

This appendix establishes the differential-geometric foundations required for the main theorems.

A.1 Fréchet Manifolds. Defines Fréchet manifolds as locally convex topological vector spaces with smooth transition maps. Notes that $\mathcal{X} = H^1(\Omega) \times H^1(\Omega; \mathbb{R}^3) \times L^2(\Omega)$ is in fact a Hilbert space (hence also a Banach space and a Fréchet space) with its product inner product. The Fréchet framing is introduced because extensions of the framework to more general field configurations (e.g., distributional fields, or fields on non-compact domains) may require Fréchet rather than Hilbert structure.

A.2 The Quotient Topology. Proves that $\mathcal{X}/\sim$ is Hausdorff under the assumption that G acts properly and isometrically. Gives the metric on $\mathcal{X}/\sim$ induced from $\mathcal{X}$. Discusses the difference between $\mathcal{X}/\sim$ (observational equivalence) and $\mathcal{X}/G$ (gauge equivalence) and when they coincide.

A.3 Tangent Bundle of $\mathcal{X}/\sim$. Defines the tangent bundle $T\mathcal{X}/\sim$ as the quotient of $T\mathcal{X}$ by the action of TG . Specifies the admissible tangent cone $\mathcal{T}_{[X]}\Gamma^0$ (Definition 6.7 of the main text) in terms of this bundle structure. Notes that when G acts freely, $T\mathcal{X}/\sim \cong T\mathcal{X}/TG$ is a vector bundle, and the tangent cone is a linear subspace.

A.4 Geodesic Convexity on $\mathcal{X}/\sim$. Defines geodesics on $\mathcal{X}/\sim$ via the quotient of geodesics on $\mathcal{X}$ (which are straight lines in the Hilbert structure). Proves that $\mathcal{A}$ is geodesically convex when its preimage in $\mathcal{X}$ is convex—which holds by the definition of $\mathcal{A}$ as the zero-sublevel set of a convex functional $\mathcal{R}$. This justifies Assumption A4 (Assumption 6.4 in the main text).

B Full Proof of the Identifiability Theorem

B.1 Functional-Analytic Setting

The proof of Theorem 6.11 relies on existence and uniqueness of minimizers of the consistency functional on $\mathcal{X}/\sim$. We introduce additional assumptions for the infinite-dimensional setting.

Assumption B.1 (Banach Realization, restated). *There exists a Banach space $\mathcal{B}$ and continuous embedding $\iota : \mathcal{X} \hookrightarrow \mathcal{B}$ such that $\mathcal{A}$ is closed and convex in $\mathcal{B}$, each Π_i extends to a bounded linear operator on $\mathcal{B}$, and $\mathcal{R}$ extends to a proper, lower semicontinuous functional on $\mathcal{B}$. In practice, $\mathcal{B} = \mathcal{X}$ itself.*

Assumption B.2 (Geodesic Convexity, restated). $\mathcal{A}$ is geodesically convex with respect to the metric induced from $\mathcal{B}$, and $\mathcal{R}$ is strictly convex along geodesics in $\mathcal{A}$:

$$\mathcal{R}(\gamma(t)) < (1-t)\mathcal{R}([X]) + t\mathcal{R}([X']) \quad \forall [X] \neq [X'] \in \mathcal{A}, t \in (0, 1).$$

Assumption B.3 (Coercivity). $f([X]) = \sum_i \|\Pi_i(X) - y_i\|^2 + \mathcal{R}(X)$ satisfies $\|X_n\|_{\mathcal{B}} \rightarrow \infty \Rightarrow f([X_n]) \rightarrow \infty$.

B.2 Existence of a Minimizer

Proposition B.4. Under the standing assumptions, f attains a minimum on $\mathcal{A}$.

Proof. Let $\{[X_n]\}$ be a minimizing sequence; coercivity gives boundedness in $\mathcal{B}$. Reflexivity of $\mathcal{B}$ (if Hilbert) yields a weakly convergent subsequence $X_{n_k} \rightharpoonup X_*$. Closedness of $\mathcal{A}$ gives $[X_*] \in \mathcal{A}$. Weak continuity of Π_i (bounded linear) and lower semicontinuity of $\mathcal{R}$ give $f([X_*]) \leq \liminf f([X_{n_k}])$. Hence $[X_*]$ is a minimizer. $\square$

B.3 Uniqueness of the Minimizer

Proposition B.5. Under the standing assumptions, the minimizer is unique.

Proof. Suppose $[X], [X']$ are distinct minimizers. By geodesic convexity, there exists a geodesic γ in $\mathcal{A}$ connecting them. The projection residual term is convex along γ (by linearity of Π_i). The regularizer is strictly convex along γ by assumption. Hence $f(\gamma(t)) < (1-t)f([X]) + tf([X']) = \inf f$ for $t \in (0, 1)$, contradicting minimality. $\square$

B.4 Passage to the Quotient

Proposition B.6. If $[X] = [X']$ in $\mathcal{X}/\sim$, then $f([X]) = f([X'])$. Conversely, distinct equivalence classes yield distinct objective values whenever $\tilde{\Pi}$ is injective.

Proof. Observational equivalence means $\Pi_i(X) = \Pi_i(X')$ for all i , so residual terms are equal; $\mathcal{R}$ is defined on equivalence classes, so regularizer terms are equal. Injectivity of $\tilde{\Pi}$ means distinct classes differ in at least one projection, so residual terms differ. $\square$

B.5 Full Proof of Theorem 6.11

Full proof. Well-definedness: by the existence and uniqueness propositions, $\mathcal{C}$ is single-valued on $\mathcal{A}$.

Fixed points = minimizers: $[X^*] = \mathcal{C}([X^*])$ iff $[X^*]$ minimizes f over $\mathcal{A}$.

Injectivity implies uniqueness: if $\tilde{\Pi}$ is injective on $\mathcal{F}$, $\mathcal{F}$ contains at most one class. By non-emptiness, $[X^*] \in \mathcal{F}$, so $|\mathcal{F}| = 1$. At $[X^*]$, all residuals are within tolerance; strict convexity prevents alternative minimizers. Hence $[X^*]$ is the unique fixed point.

Non-injectivity implies non-uniqueness: if $\tilde{\Pi}$ is not injective on $\mathcal{F}$, distinct $[X], [X'] \in \mathcal{F}$ have identical projections. Both are candidates for the minimizer. If $\mathcal{R}([X]) = \mathcal{R}([X'])$, strict convexity fails (contradiction). If $\mathcal{R}([X]) \neq \mathcal{R}([X'])$, the larger-regularizer point is not a minimizer—but then it is not in $\mathcal{F}$ as defined, contradicting the assumption. Either way we reach a contradiction with uniqueness.

□

□

C Full Proof of the Dynamical Disambiguation Theorem

C.1 Measure-Theoretic Framework

Assumption C.1 (Polish Structure). $\mathcal{A}$ with the metric induced from $\mathcal{B}$ is a Polish space (complete, separable metric).

Definition C.2 (Markov Kernel, precise). $T_\Delta : \mathcal{A} \rightarrow \mathcal{P}(\mathcal{A})$ is a Markov kernel: for each $[X]$, $T_\Delta([X], \cdot)$ is a Borel probability measure on $\mathcal{A}$; for each Borel $B \subseteq \mathcal{A}$, $[X] \mapsto T_\Delta([X], B)$ is measurable.

C.2 Convergence to the Deterministic Generator

Assumption C.3 (Generator Convergence, precise). For all bounded continuous $\varphi : \mathcal{A} \rightarrow \mathbb{R}$:

$$\int \varphi(X') T_\Delta([X], dX') = \varphi([X] + \Delta F([X]; \theta)) + O(\Delta^{3/2}).$$

This is weak convergence $T_\Delta([X], \cdot) \Rightarrow \delta_{[X] + \Delta F([X]; \theta)}$ at rate $\Delta^{3/2}$, corresponding to a diffusion with noise variance $O(\Delta)$.

C.3 Stability of Trajectory Separation Under Noise

Definition C.4 (Uniform Separation Margin). A pair $[X], [X'] \in \Omega_{\text{obs}}$ is uniformly trajectory-separated if there exist $\tau > 0$ and $\delta > 0$ such that

$$\left\| \tilde{\Pi}(\Phi_F^\tau([X])) - \tilde{\Pi}(\Phi_F^\tau([X'])) \right\| \geq \epsilon + \delta.$$

Lemma C.5. *Under generator convergence, uniformly trajectory-separated pairs remain distinguishable under stochastic evolution with probability $\geq 1 - C\Delta/\delta^2$.*

Proof. By Chebyshev applied to the $O(\Delta)$ noise variance, stochastic trajectories deviate from deterministic by $O(\sqrt{\Delta})$ with high probability. For deviation $\delta/2$: $\mathbb{P}(\|X_\tau^\Delta - X_\tau\| > \delta/2) \leq C\Delta/\delta^2$. With margin δ , the two stochastic trajectories remain separated by $\geq \epsilon$ with probability $\geq 1 - 2C\Delta/\delta^2$. $\square$

C.4 Full Proof of Theorem 7.10

Full proof. Let $[X], [X'] \in \Omega_{\text{obs}} \cap \Gamma^0$ be distinct and uniformly trajectory-separated at time τ .

Step 1: Both lie on integral curves of $F(\cdot; \hat{\theta})$ by definition of Γ^0 .

Step 2: At time τ , their projections differ by $\geq \epsilon + \delta > \epsilon$.

Step 3: By the noise lemma, stochastic evolutions remain separated with high probability for small Δ .

Step 4: At time τ , the true observation $y(\tau)$ has $\|y_i(\tau) - \Pi_i(X^*(\tau))\| \leq \epsilon_i$. At most one of the two separated candidates lies within ϵ_i of $y_i(\tau)$.

Step 5: The true state $[X^*(\tau)] \in \Omega_{\text{obs}}(y(\tau))$ by non-emptiness applied at time τ . Hence the false candidate is eliminated.

Step 6: Applying to all pairs eliminates all but $[X^*]$. Hence $|\mathcal{F}| = 1$. $\square$ $\square$

D Fisher Information Geometry of the Feasible Set

[PRELIMINARY OUTLINE — NOT YET DRAFTED]

This appendix develops the Riemannian geometry of $\mathcal{F}$ induced by the noise models $\{\mathcal{N}_i\}$ and connects it to the entropy identification of Section 2.

D.1 The Fisher Information Metric. Defines the Fisher information metric on $\mathcal{X}/\sim$ induced by the family of observation distributions $\{p_i(\cdot \mid [X]) = \mathcal{N}(\Pi_i(X), \Sigma_i)\}_{i \in \mathcal{I}}$. For Gaussian noise models with covariance Σ_i , the Fisher metric at $[X]$ is:

$$g_F([X])(u, v) = \sum_i \langle D\Pi_i(u), \Sigma_i^{-1} D\Pi_i(v) \rangle_{\mathcal{Y}_i}$$

where $D\Pi_i$ is the derivative of Π_i at $[X]$. This is a semi-Riemannian metric on $\mathcal{X}/\sim$: it may be degenerate in directions where all projections have zero derivative.

D.2 Volume of the Feasible Set. Defines $\text{Vol}_{\text{Fisher}}(\mathcal{F})$ as the Riemannian volume of $\mathcal{F}$ under g_F . Shows that $\text{Vol}_{\text{Fisher}}(\mathcal{F}) \propto \prod_i \epsilon_i^{n_i}$ where n_i is the number of degrees of freedom closed by Π_i not already closed by $\{\Pi_1, \dots, \Pi_{i-1}\}$ (the marginal contribution of each sensor). As $\epsilon_i \rightarrow 0$ for all i , $\text{Vol}_{\text{Fisher}}(\mathcal{F}) \rightarrow 0$, corresponding to closure.

D.3 Connection to Entropy. Proves the identification $\log \text{Vol}_{\text{Fisher}}(\mathcal{F}) \leftrightarrow \log \dim H^0(\mathcal{U}, \mathcal{S}_{\Pi}) \leftrightarrow S_{\text{RSVP}}(X^*)$ stated in Section 2. The first identification uses the fact that $\dim H^0$ counts linearly independent global sections, which in the Gaussian regime scales with the volume of the feasible set. The second uses the RSVP entropy field equation and the fact that at the fixed point $[X^*]$, S_{RSVP} is the integral of local entropy production over Ω , which is proportional to the log-volume of remaining uncertainty.

D.4 Reconstruction Efficiency. Defines the *reconstruction efficiency* of a sensor network as the rate of $\text{Vol}_{\text{Fisher}}(\mathcal{F})$ decrease per unit of added sensing capacity (measured in total Fisher information $\sum_i \|\Sigma_i^{-1}\|_{\text{op}}$). This gives a principled criterion for sensor placement: the next sensor should be placed where it maximizes the decrease in $\text{Vol}_{\text{Fisher}}(\mathcal{F})$, which is determined by the spectrum of the Fisher metric in the remaining unclosed directions.

D.5 Convergence Rate of the Refinement Loop. Bounds the convergence rate of the refinement loop (Section 8) in terms of the eigenspectrum of g_F at $[X^*]$. The refinement loop is a gradient descent on f ; its convergence rate is determined by the condition number $\kappa = \lambda_{\max}/\lambda_{\min}$ of the Hessian of f at $[X^*]$, which is related to the eigenvalues of g_F at $[X^*]$. In particular, directions with small Fisher information (loosely constrained by sensors) correspond to directions with small Hessian eigenvalues and hence slow convergence. This gives a formal account of why dense, multi-modal sensor networks accelerate convergence to closure.

E Operator Class Library

[PRELIMINARY OUTLINE — NOT YET DRAFTED]

This appendix formally specifies the named transition operators in the operator class library introduced in Section 7.

E.1 Structure of the Library. The library consists of K named process generators $\{F_k\}_{k=1}^K$, each a learned vector field on $\mathcal{A}$ parametrized by $\theta_k \in \Theta_k$. The full dynamics are $F([X]; \theta) = \sum_k \alpha_k([X]) F_k([X]; \theta_k)$ with $\alpha_k \geq 0$, $\sum_k \alpha_k = 1$. Each F_k is learned from before/after pairs in which process k is dominant, identified by its characteristic sensor signature.

E.2 Ecological Process Operators. Specifies the following operators with their characteristic sensor signatures:

Forest succession (F_{succ}): slow increase in LiDAR canopy height, gradual shift in RGB reflectance from grass-like to canopy-like, decrease in audio spectral entropy (bird diversity changes with succession stage). Time scale: years to decades.

Disturbance and recovery (F_{dist}): rapid decrease in LiDAR height and RGB NDVI, followed by slow recovery. Audio signature: absence of characteristic species. Time scale: event-driven (days), recovery (years).

Hydrological dynamics (F_{hydro}): correlated changes in infrared (soil moisture affects surface temperature), LiDAR (inundation changes geometry), and audio (water flow sounds). Time scale: seasonal to event-driven.

Phenological cycling (F_{pheno}): regular seasonal oscillations in RGB (green-up/senescence), infrared (latent heat flux changes), and audio (bird song seasonality). Time scale: annual.

E.3 Civilizational Process Operators. Specifies:

Urban expansion (F_{urban}): progressive increase in LiDAR building density and height, change in RGB to impervious surface reflectance, increase in audio (traffic, machinery). Time scale: years to decades.

Infrastructure decay (F_{decay}): gradual change in LiDAR surface roughness, spectral shift in RGB, increase in audio entropy (structural noise). Time scale: years.

Agricultural cycling (F_{agri}): regular oscillations in RGB (crop growth stages), infrared (ET changes), LiDAR (canopy height variation), audio (machinery activity). Time scale: seasonal.

Demographic change (F_{demo}): correlated changes in audio activity levels, RGB night-light proxies, and infrastructure use patterns. Time scale: years.

E.4 Learning Protocol. Specifies how each F_k is learned from before/after pairs: (a) identify pairs where process k is the dominant change by classifying the observation delta against the library; (b) fit θ_k by minimizing the discrepancy between the predicted forward image $\Pi_i(\Phi_{F_k}([X_b]))$ and the actual after-observation y_a over training pairs; (c) validate by held-out pairs and by checking that $\Theta_{(y_b, y_a, \tau)} \cap \Theta_k$ is non-empty for pairs labeled as process k .

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